

Front Wing Repair

WORK IN PROGRESS

Root Cause Analysis: Front Wing Failure

After reaching around 150lbs of downforce during wind tunnel testing (___ mph airspeed), the front wing ribs deflected enough to strain the bond between the outer sections of the mainplane, causing a fracture in the filler, along with some separation/fracture in the leading edge. As the mainplane ribs are only constrained by their hardpoints to the front wing struts, which in turn is mounted to the monocoque, it is reasonable to assume that when loading both sides of the mainplane downwards, there is nothing to react the moment incurred besides the stiffness of the rib and strut, which leads to most of the plies /filler taking the load and deflecting.



Figure 1: Relief cuts made in front wing, possible point failure



Figure 2: Small crack located in the filler, rightmost of the left center mainplane rib



Figure 3: Separation at the leading edge between each "half" of the mainplane. When loaded downwards, the gap increases before returning back to depicted state.



Figure 4: Noticeable crack in the filler located at the right center mainplane rib

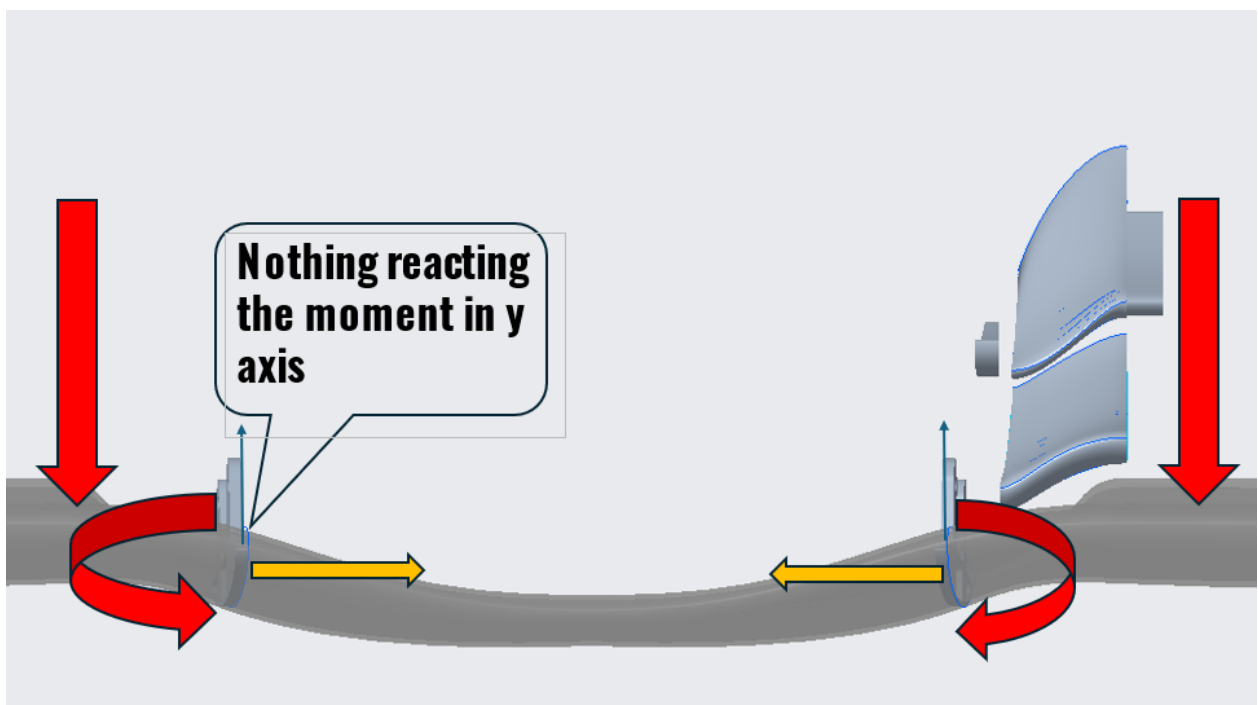
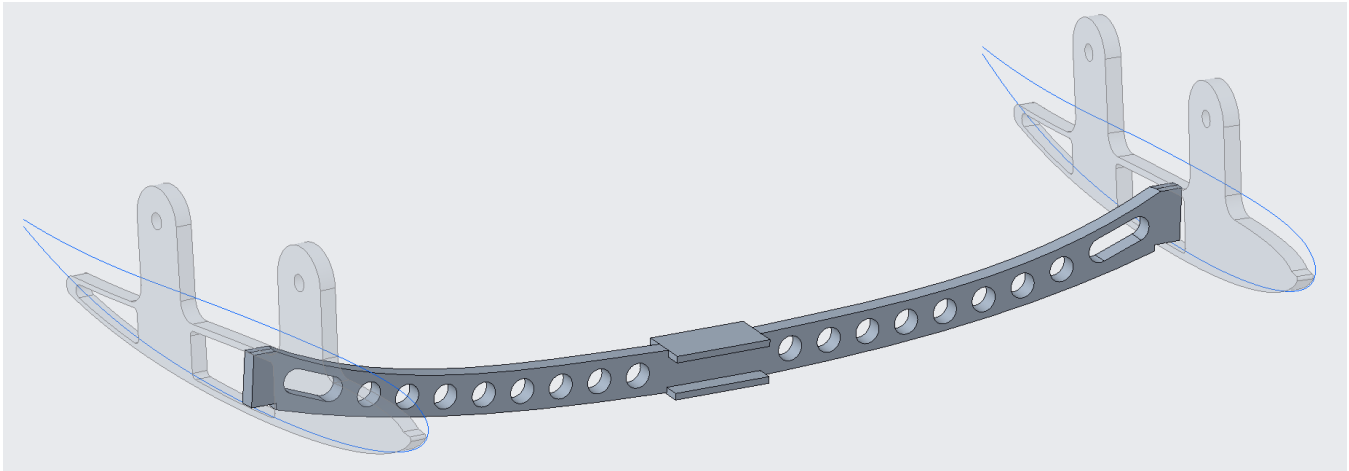


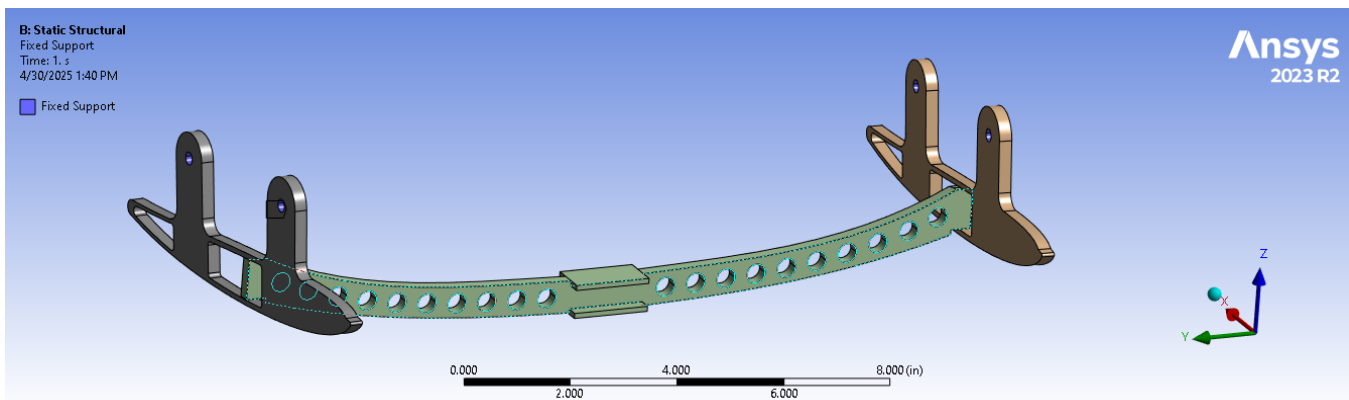
Figure 5: Artist's rendition on cause for failure

Spar Design

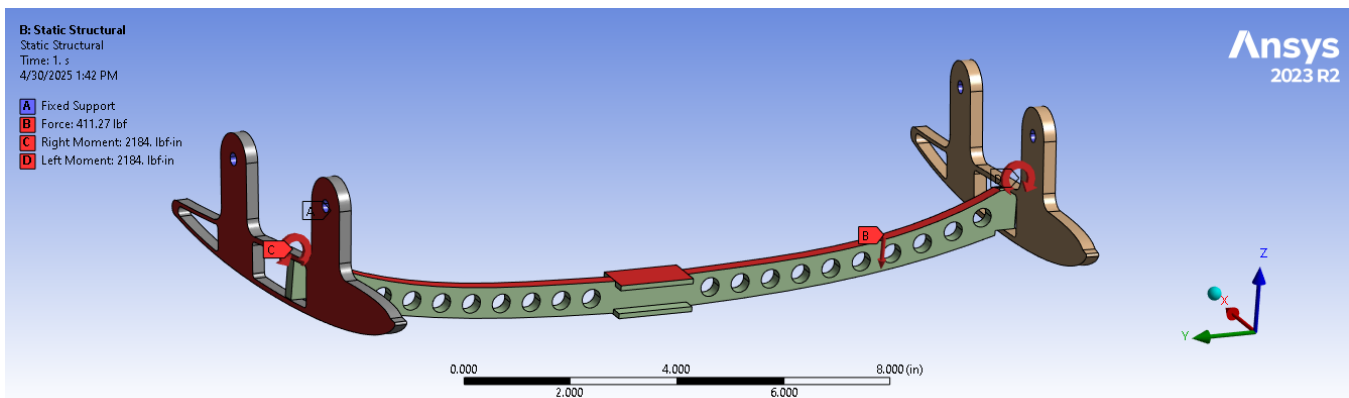
Given the usage of a 1-core-1 panel for the upper camber, there is fairly little space for the spar to exist in, in addition to the curve of the mainplane profile. The proposed solution is to have a curved spar running down the span of the central mainplane situated at the CoP, connected to each rib either through an epoxy bond or bolted joint.



FEA setup

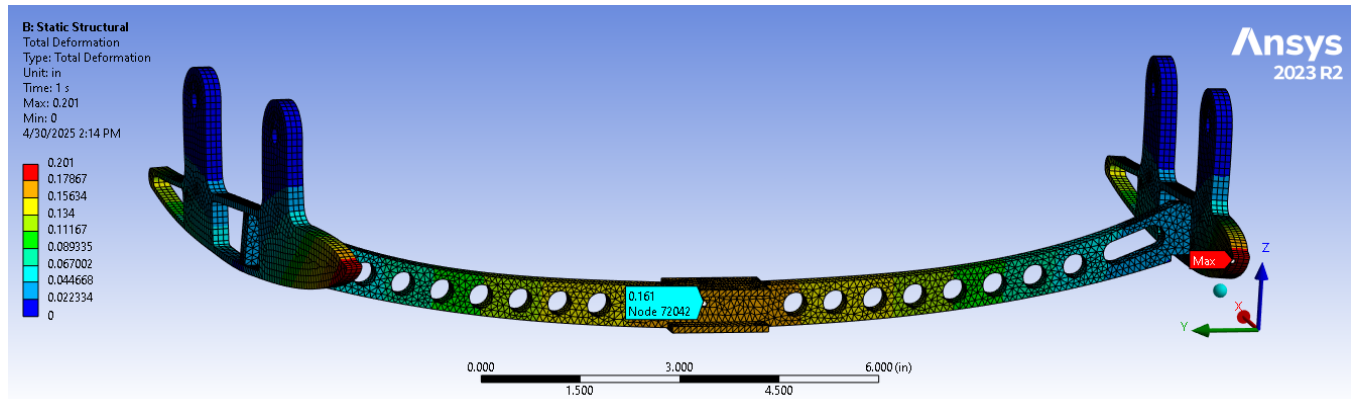


Supports



applied forces

Results



jonny feedback:

fixed supports on rib tabs anti-conservative, model as pin since struts have bad torsional stiffness set upper and lower bound for deflection

better plan on joining method, cad to irl relations, plan to reassemble lower mainplane surface

Joining Method

Repair Steps

- Waterjet spar
- weld plates to middle section
- Cut open fw
- match drill spar and rib
- tap and bolt spar in
- reseal fw